Power and Precision in Balanced vs. Unbalanced Case–Control Studies

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`https://github.com/ranlib/power_and_precision_calculation_for_case_control_studies`

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1 Overview

We compare study designs in terms of:

- **Statistical power**
- **Precision (confidence interval width)**

Key message: Increasing controls yields diminishing returns for both power and precision, while increasing cases is far more effective.

2 Model

$$p_1 = \frac{OR \cdot p_0}{1 - p_0 + OR \cdot p_0}$$
$$\mathrm{Var}(\log OR) = \frac{1}{n_1 p_1} + \frac{1}{n_1(1-p_1)} + \frac{1}{n_0 p_0} + \frac{1}{n_0(1-p_0)}$$

3 Parameters

- $OR = 2$
- $p_0 = 0.2$
- $\alpha = 0.05$
- $z_{0.975} = 1.96$

4 Results

4.1 Power

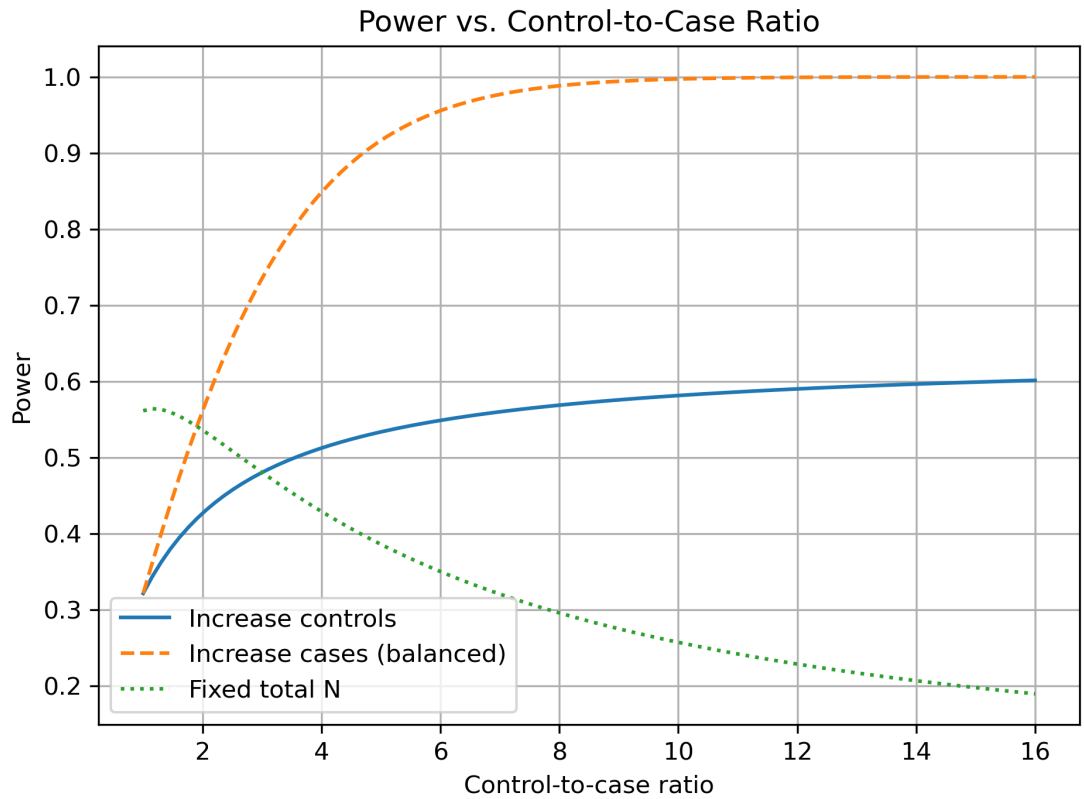

Figure 1: Power vs. control-to-case ratio

4.2 Confidence Interval Width

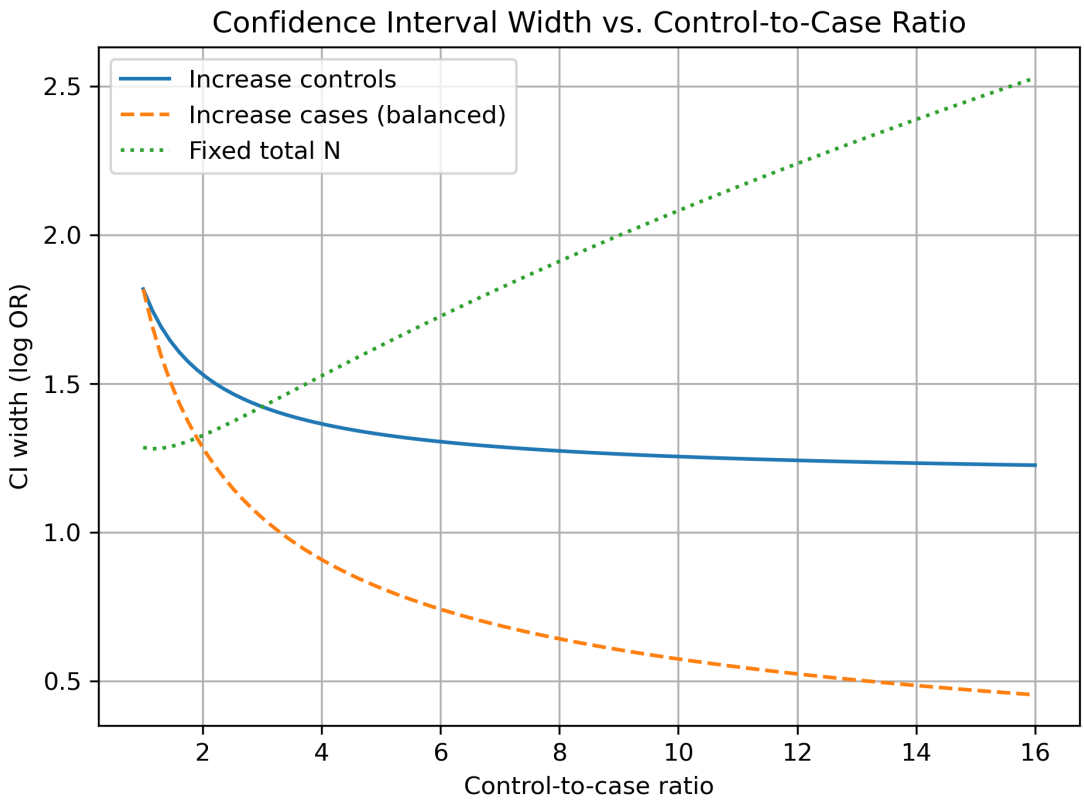

Figure 2: Power vs. control-to-case ratio

5 Interpretation

- Increasing controls reduces CI width initially, but quickly plateaus
- Increasing cases leads to continuous improvement in precision
- For fixed total sample size, balanced designs minimize CI width

6 Conclusion

- Power and precision are primarily driven by the number of cases
- Additional controls provide diminishing returns
- Balanced designs are optimal when feasible

7 When Extra Controls Do Help Meaningfully

While increasing the number of controls in a case–control study typically yields diminishing returns in statistical power, there are important scenarios in which adding controls can still provide meaningful benefits.

7.1 Low Baseline Exposure Probability

When the exposure probability in controls (p_0) is very small, the variance contribution from the control group becomes large:

$$\mathrm{Var}(\log OR) = \frac{1}{n_1 p_1} + \frac{1}{n_1(1-p_1)} + \frac{1}{n_0 p_0} + \frac{1}{n_0(1-p_0)}$$

In such settings, increasing the number of controls can substantially reduce variance and improve power, even beyond the usual 4:1 control-to-case ratio rule of thumb.

7.2 Rare Outcomes with Limited Cases

In many real-world studies (e.g., rare diseases), the number of cases is fixed and difficult to increase. In this situation:

- Increasing cases is not feasible
- Increasing controls is often inexpensive and practical

Although gains in power plateau, additional controls still provide incremental improvements and may be the only viable strategy.

7.3 Improved Precision of Effect Estimates

Even when power gains are small, increasing the number of controls reduces the width of the confidence interval:

$$\text{CI width} \propto \sqrt{\mathrm{Var}(\log OR)}.$$

This is important when:

- Estimation (effect size precision) matters more than hypothesis testing
- Results are used for downstream modeling or meta-analysis

7.4 Control Reuse and External Data Sources

In some designs, controls can be obtained at very low cost:

- Public databases
- Biobanks
- Previously collected cohorts

When controls are effectively "free," it can be advantageous to include many of them, even if statistical gains are modest.

7.5 Adjustment for Confounding

Larger control groups can improve:

- Stability of regression models
- Estimation of covariate effects
- Matching or stratification procedures

This is especially relevant in high-dimensional settings or when adjusting for many confounders.

7.6 Summary

Although balanced designs are statistically optimal for a fixed total sample size, increasing the number of controls can still be beneficial when:

- Exposure is rare
- Cases are limited
- Precision (not just power) is important
- Controls are inexpensive or readily available

Thus, the choice of control-to-case ratio should reflect not only statistical efficiency, but also practical constraints and study objectives.

A Interpretation of Model Parameters

This appendix describes the meaning of the parameters used in the power and sample size calculations for the case–control study.

- **Odds Ratio** ($OR = 2$)
  The odds ratio represents the strength of association between the exposure and the outcome. An odds ratio of 2 indicates that the odds of exposure among cases are twice the odds of exposure among controls. Formally,

  $$OR = \frac{\frac{p_1}{1-p_1}}{\frac{p_0}{1-p_0}},$$

  where p_1 is the exposure probability among cases and p_0 is the exposure probability among controls.

- **Control Exposure Probability** ($p_0 = 0.2$)
  This is the assumed prevalence of the exposure in the control group, which serves as a proxy for the exposure prevalence in the underlying source population. The value of p_0 determines the variance of the estimated odds ratio and therefore directly influences statistical power.

- **Significance Level** ($\alpha = 0.05$)
  The significance level α is the probability of a Type I error, i.e., rejecting the null hypothesis when it is true. A two-sided $\alpha = 0.05$ corresponds to a 5% chance of falsely declaring an association when none exists.

- **Critical Value ($z_{0.975} = 1.96$)**
  The value $z_{0.975}$ is the 97.5th percentile of the standard normal distribution. It is used in two-sided hypothesis testing and confidence interval construction with $\alpha = 0.05$, since

  $$z_{1-\alpha/2} = z_{0.975} \approx 1.96.$$

  This value defines the rejection threshold for the test statistic and determines the width of the confidence interval.

Together, these parameters determine the expected distribution of the log odds ratio estimator, its variance, and the resulting statistical power and confidence interval width in the study design.

This topic is discussed in [1].

References

[1] Jannatul Ferdous et al. The rise to power of the microbiome: power and sample size calculation for microbiome studies. *Mucosal Immunology*, 15(5):1061–1074, 2022.